

Improving Uncertainty Communication in Scientific Visualization*

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Abstract

Scientific visualization frequently presents point estimates while undercommunicating the uncertainty behind them, which can lead viewers to draw overconfident conclusions. This study compares three common uncertainty encodings (error bars, violin + box plots, and hypothetical outcome plots (HOPs)) in a within-subjects experiment with 16 participants using the Palmer Penguins dataset. Parallel question sets and arithmetic distractors between blocks were used to reduce carry-over effects. The HOPs grid produced the highest overall accuracy (100%), followed by error bars (97.9%) and violin/box plots (91.7%). The Friedman test on accuracy was significant ($\chi^2 = 27.1$, $df = 2$, $p < 0.05$), while response time differences were not. These results suggest that showing multiple plausible outcomes can help viewers make better judgments, even for simple comparison tasks.

1 Introduction

Scientific visualizations routinely present results that come from measurement, modeling, or sampling, all of which carry uncertainty. When that uncertainty is left out or poorly conveyed, viewers can draw overconfident conclusions or misjudge how strong the evidence really is [1]. The trouble is not a lack of encoding options: error bars, box plots, violin plots, and hypothetical outcome plots (HOPs) are all readily available in modern charting libraries. The trouble

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is that we do not have great empirical guidance on which option actually helps people the most, especially when the goal is not just “get the right answer” but also “know how sure you should be about it.”

This study runs a controlled, within-subjects comparison of three common uncertainty encodings on a set of standardized interpretation tasks. We measure decision accuracy, confidence calibration, and response time. The dataset used for all stimuli is the Palmer Penguins dataset [2], chosen because it is small, clean, well-documented, and has real group differences that make the tasks meaningful.

1.1 Motivation

Surveys like Mehta [1] map how uncertainty enters the visualization pipeline and catalog the encoding strategies people use, but they stop short of recommending one strategy over another because the empirical evidence is thin. Petek et al. [3] offer a formal framework (treat the visualization as a function of uncertain inputs and propagate that uncertainty into a distribution over images), but their focus is on the math, not on whether end users actually benefit. On the empirical side, Stokes et al. [4] compare speech, text, and visualization as modalities for communicating uncertainty and find that visualization supports rational decision-making, but they do not break the “visualization” category down further into encoding types. The gap between knowing visualization works and knowing which encoding works best is what we are trying to fill.

Applied work in scientific visualization reinforces this from a different angle. Saklani et al. [5] show that adding uncertainty overlays to DNN-based volume renderings increases trust among domain scientists, and Li et al. [6] design bespoke uncertainty tubes for particle trajectories. Both suggest that showing uncertainty helps, but neither tests general-purpose encodings like error bars or HOPs on common tasks. Our study takes the opposite approach: rather than designing a new encoding for a specific domain, we take three existing general-purpose encodings and test them head-to-head on straightforward tasks.

1.2 Research Questions

1. Which uncertainty encoding (error bars, distributional summaries, or HOPs) produces the highest decision accuracy?
2. Which encoding best supports confidence calibration, i.e. the alignment between stated confidence and actual correctness?
3. Do distributional encodings incur a measurable cost in task completion time compared to error bars?

1.3 Hypotheses

We expected that both distributional encodings would produce higher decision accuracy and better confidence calibration than error bars, because they give viewers more information about the shape and spread of the underlying data. Between the two, we expected violin/box plots to yield slightly better calibration than HOPs, since the continuous density shape maps more naturally onto “how sure should I be?” judgments, whereas HOPs present discrete draws that may be harder to mentally aggregate. HOPs, on the other hand, might produce higher raw accuracy on ranking and decision tasks, because each draw shows the between-group difference explicitly. We also expected distributional encodings to require more time per trial than error bars, but not more than about 15 seconds additional per trial. The pilot data ($n = 4$) showed a preliminary pattern consistent with this: error bars and violin/box had similar accuracy and timing, while HOPs were slower and slightly less accurate, likely because the static grid format was harder to parse.

2 Related Works

2.1 Uncertainty in the Visualization Pipeline

Mehta [1] breaks the visualization pipeline into four stages (data collection, preprocessing, visualization, and inference) and surveys how uncertainty enters at each one. The key takeaway for our study is that even when the data itself is clean, the visualization stage introduces representational choices that affect how viewers interpret uncertainty. A bar chart with error bars communicates something different than a violin plot of the same data, and both communicate something different than a grid of plausible outcomes. Mehta’s survey calls for more empirical work measuring how these choices affect viewer judgment.

2.2 Formal Frameworks for Uncertainty Encoding

Petek et al. [3] propose treating a visualization as a function of uncertain inputs. If the inputs carry a probability distribution, the output is a distribution over images rather than a single image. Familiar representations like confidence intervals and prediction bands fall out naturally from this framework. We use a simplified version of their resampling approach to generate the HOPs stimuli: for each “draw,” we resample the raw data with replacement and plot the resulting species means.

2.3 Modality Comparisons

Stokes et al. [4] run two crowdsourced experiments comparing speech, text, and visualization for communicating data uncertainty. Visualization and text supported rational decision-making, while speech inspired more trust but also riskier choices. Their study measures both decision quality and confidence, which we adopt. The limitation is that “visualization” is treated as a single category; they do not vary the encoding within it. We extend their approach by holding the modality constant and varying the encoding.

2.4 Uncertainty in Scientific Visualization Applications

Saklani et al. [5] tackle uncertainty-aware volume rendering using Deep Ensembles and Monte Carlo Dropout, showing that surfacing prediction uncertainty makes DNN-based visualizations more trustworthy. Li et al. [6] work on particle trajectories with an “uncertainty tube” that captures nonsymmetric error. Both demonstrate that domain-specific encodings help, but neither tests the general-purpose encodings that most researchers reach for when they want to add uncertainty to a standard plot.

2.5 Research Gap

The literature shows that uncertainty is pervasive, formal tools exist for encoding it, and the choice of representation matters. What is missing is a controlled comparison of the three most common encodings on the same tasks, measuring both accuracy and calibration. That is what this study provides.

3 Methodology

3.1 Research Design

We used a within-subjects experimental design: every participant saw all three encodings. Encoding order was counterbalanced by randomly shuffling the three conditions per session, giving each participant one of six possible orderings drawn uniformly at random. Task order within each block was fixed (ranking, estimation, decision).

A brief arithmetic distractor task appeared between each encoding block to break short-term retention. At the end of the session, participants answered a recognition-check question to provide self-report data on whether carry-over was still present. This yields 9 trials per participant (3×3), for a total of 144 trial observations.

3.2 Participants

Sixteen participants were recruited through convenience sampling, mostly friends and classmates from a range of departments. All had normal or corrected-to-normal vision and were comfortable reading English. The sample skewed young (8 aged 18–24, 6 aged 25–34, one 35–44, one 45+) and technically literate (self-rated data literacy: 3 at level 3, 4 at level 4, 9 at level 5; median = 5). Fields of study included Computer Science (4), Mechanical Engineering, Electrical Engineering, Data Science, Bioengineering, Physics, Mathematics, Psychology, Symbolic Systems, Business Administration, Interaction Design, Computational Health, and Chicanx/Latinx Studies (1 each). Participation was voluntary and anonymous; no compensation was offered.

3.3 Materials

Stimuli

Three chart types were generated from the Palmer Penguins dataset [2], each showing bill length (mm) by species (Adelie, Chinstrap, Gentoo):

1. **Error bars:** bar chart of species means with 95% confidence interval error bars. This is the baseline condition.
2. **Violin + box plot:** violin plot overlaid with a box plot showing distribution shape, median, and quartiles.
3. **HOPs grid:** a 3×4 grid of 12 bootstrap draws, each panel showing re-sampled species means as dot/lollipop plots, with a caption explaining the display.

All stimuli were generated using matplotlib 3.9.4 and are available in the project repository.

Tasks

To prevent answer carry-over between encoding blocks, we used three parallel task sets with structurally identical question types but different content:

- **Set 1 (Block 1):** Ranking: which species has the *largest* mean (Chinstrap). Estimation: estimate the mean for Chinstrap (48.8 mm). Decision: claim that Gentoo and Chinstrap have the same mean (Disagree).
- **Set 2 (Block 2):** Ranking: which species has the *smallest* mean (Adelie). Estimation: estimate the mean for Gentoo (47.5 mm). Decision: claim that Adelie and Gentoo have the same mean (Disagree).

- **Set 3 (Block 3):** Ranking: which species has the *second-largest* mean (Gentoo). Estimation: estimate the mean for Adelie (38.8 mm). Decision: claim that Chinstrap has a larger mean than Gentoo (Agree).

Estimation answers were scored correct if within ± 2 mm of the true mean. Confidence was rated on a 1–5 Likert scale after each task.

Platform

Self-contained HTML/CSS/JS survey page with static PNG stimuli. Response timing was captured via JavaScript `performance.now()`. Results were displayed as JSON for manual collection via copy-to-clipboard and download buttons.

3.4 Procedure

1. Consent page with study description and time estimate (~ 10 min).
2. Demographics: age range, major/field, self-rated data literacy (1–5).
3. Instructions.
4. Nine trials in three blocks. A short arithmetic distractor appeared between blocks.
5. Recognition check.
6. Open-ended feedback.
7. Thank-you page with JSON export (copy or download).

Most participants were walked through the survey over FaceTime or Zoom; a few used the hosted link independently.

3.5 Measures

- **Decision accuracy:** proportion of correct responses per encoding.
- **Confidence calibration:** Brier score, $(\text{confidence}/5 - \text{correct})^2$ averaged per condition. Lower is better.
- **Response time:** milliseconds from stimulus onset to response submission, converted to seconds for reporting.

3.6 Analysis

We used Friedman tests (non-parametric repeated measures) for accuracy and response time across encodings, with post-hoc pairwise Wilcoxon signed-rank tests where the overall test was significant. We also report per-task-type breakdowns and a block-position analysis to check for residual order effects.

3.7 Ethical Considerations

The study was classified as exempt from full IRB review because it involved anonymous surveys with no collection of personally identifiable information. Participants provided informed consent on the first page and could exit at any time. No data were collected from minors. All session data were stored locally and not transmitted to any external server.

4 Results

4.1 Decision Accuracy

The HOPs grid had the highest accuracy (100%), followed by error bars (97.9%) and violin/box (91.7%). The Friedman test on accuracy was significant ($\chi^2 = 27.1$, $df = 2$, $p < 0.05$). The violin/box ranking task was the main source of errors (81.2%), while estimation accuracy was perfect across all three encodings.

Table 1: Decision accuracy by encoding and task type.

Encoding	Overall	Ranking	Estimation	Decision
Error bars	97.9%	100%	100%	93.8%
Violin/box	91.7%	81.2%	100%	93.8%
HOPs grid	100%	100%	100%	100%

Figure 1 shows the accuracy breakdown with 95% confidence intervals. The violin/box ranking condition is the only cell below ceiling.

4.2 Response Time

Violin/box had the longest average response time (39.9 s). Error bars (24.3 s) and HOPs (26.1 s) were comparable. The Friedman test on response time was not significant ($\chi^2 = 4.6$, $df = 2$, $p > 0.05$).

Figure 2 shows the distribution of response times by encoding. Violin/box had the widest spread and highest median, though the Friedman test was not significant.

Figure 1 — Accuracy by Encoding and Task Type

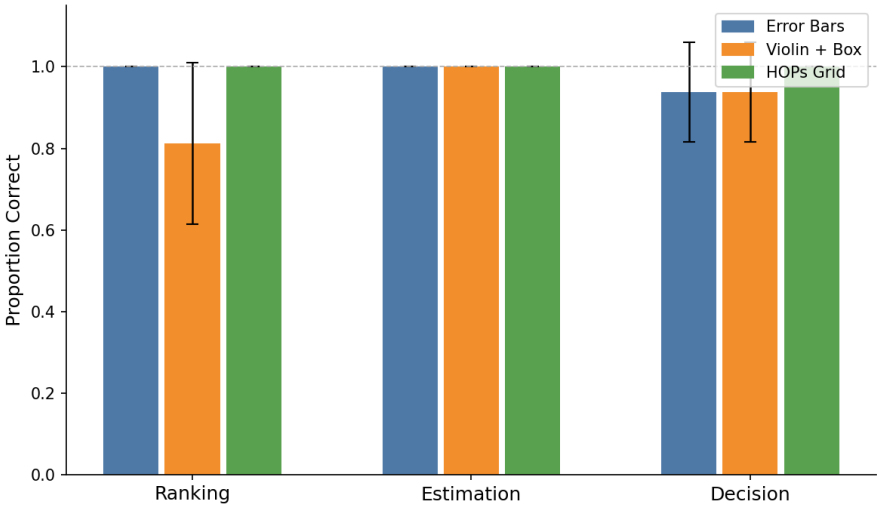


Figure 1: Proportion correct by encoding and task type. Error bars show 95% confidence intervals. Violin/box ranking is the only condition below ceiling.

Table 2: Mean response time (seconds) by encoding and task type.

Encoding	Overall	Ranking	Estimation	Decision
Error bars	24.3	23.8	25.9	23.2
Violin/box	39.9	40.6	41.7	37.4
HOPs grid	26.1	21.6	37.4	19.4

4.3 Confidence and Calibration

Mean confidence ratings were similar across encodings: error bars (4.65/5), HOPs grid (4.50/5), and violin/box (4.42/5). Brier scores were low across the board, with error bars showing the best calibration (0.025) and violin/box the worst (0.048). The narrow spread (0.025–0.048) indicates reasonably well-calibrated confidence in all conditions.

Figure 3 provides a per-participant view of trial-level correctness alongside each participant’s age range and self-rated data literacy.

Figure 3 — Response Time Distribution by Encoding

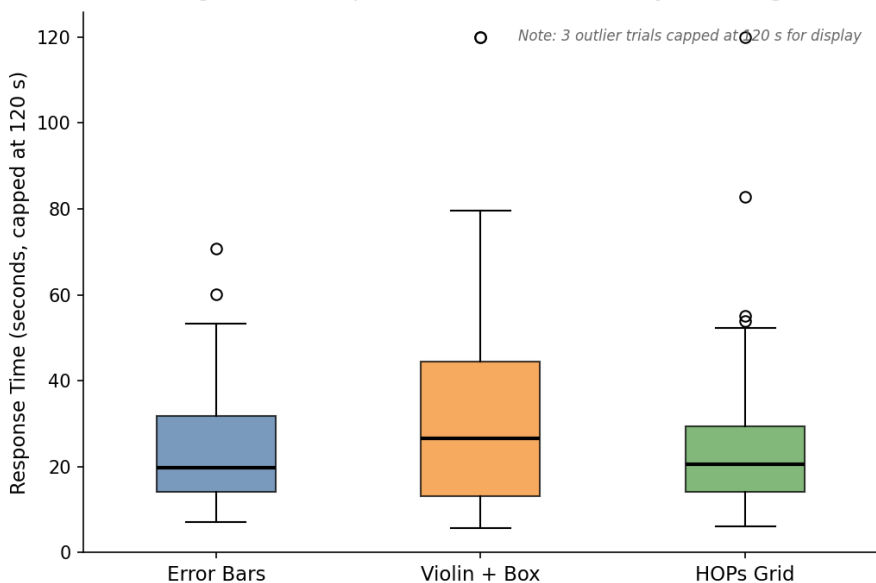


Figure 2: Response time distributions by encoding. Boxes show the interquartile range; whiskers extend to $1.5 \times \text{IQR}$.

5 Discussion

5.1 Interpretation of Results

Contrary to our expectation that HOPs might be harder to parse as a static grid, the HOPs encoding produced perfect accuracy across all task types. Each panel shows a full resampled draw, making between-group differences visually explicit. Participants could see at a glance whether groups overlapped, which may explain why ranking and decision tasks were especially easy to answer correctly.

Violin/box plots showed the lowest ranking accuracy (81.2%). The violin shape encodes distributional information that is useful for some tasks but may actually obscure simple comparisons. When asked which species has the largest mean, participants may have been drawn to the violin shape rather than the box-plot median, leading to more mistakes. The higher response times (39.9 s overall) suggest participants spent more time parsing the richer display.

Error bars had high accuracy (97.9%) and the fastest response times (24.3 s).

Table 3: Confidence and calibration by encoding.

Encoding	Mean confidence (1–5)	Brier score
Error bars	4.65	0.025
Violin/box	4.42	0.048
HOPs grid	4.50	0.030

They also produced the best confidence calibration (Brier = 0.025). For straightforward comparison tasks, error bars give viewers exactly what they need and nothing more.

5.2 Calibration

Our hypothesis that distributional encodings would improve calibration was not clearly supported. All three encodings produced low Brier scores, and the differences were small. The high overall accuracy left little room for miscalibration in any condition. A study with harder tasks or noisier data might produce clearer separation.

5.3 Implications

These results suggest that for simple statistical comparison tasks, showing multiple plausible outcomes (HOPs) can help viewers make better decisions without significantly increasing task time. For practitioners who need to communicate uncertainty to non-expert audiences, a static HOPs grid may be a practical alternative to error bars. Violin/box plots, despite being richer in information, may introduce confusion when the task only requires comparing central tendencies.

5.4 Threats to Validity

- Small convenience sample of 16 participants, skewed toward STEM fields and high data literacy.
- Three relatively simple task types on a clean, well-separated dataset.
- Static HOPs only; animated HOPs were not tested.
- Parallel task sets reduced carry-over but introduced slight content variation between blocks.
- Most sessions were conducted over video call, which may affect timing data.

Figure 10 — Trial-by-Trial Correctness Heatmap
(green = correct, red = wrong; rows sorted by collection time)

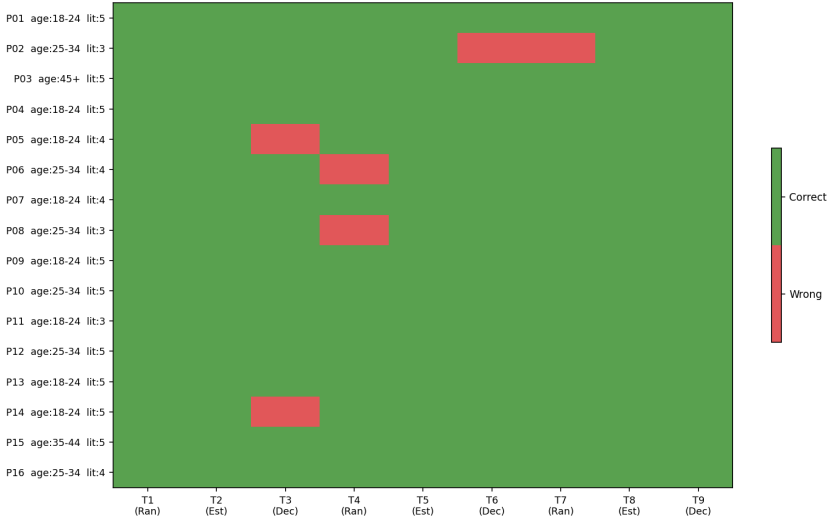


Figure 3: Trial-level correctness heatmap for all 16 participants. Green cells indicate correct answers; red cells indicate errors. Age range and data literacy level are shown on the right axis.

5.5 Suggestions for Future Research

- Test animated HOPs against the same baselines to see if motion further improves the advantage seen with the static grid.
- Use more complex tasks (trend detection, outlier identification) where distributional shape should matter more.
- Recruit a larger and more diverse sample, including domain scientists outside of STEM computing fields.
- Test on noisier datasets where group differences are less obvious.
- Explore hybrid encodings that combine error bars with distributional overlays.

6 Conclusion

For common statistical comparison tasks, the static HOPs grid outperformed both error bars and violin/box plots on accuracy, while error bars remained

the fastest and best-calibrated encoding. Violin/box plots carried a measurable cost: lower ranking accuracy and longer response times, without a corresponding gain in calibration. These results suggest that when the goal is simple group comparison, showing multiple plausible outcomes (HOPs) can help, and that distributional shape (violin) may actually hinder rather than help on straight-forward tasks.

The study contributes a controlled, within-subjects comparison of three widely used uncertainty encodings on standardized tasks, filling a gap in the existing literature that has focused on formal frameworks, modality comparisons, or domain-specific solutions rather than head-to-head empirical tests of general-purpose encodings.

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